THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** simplified representation	Simplification.
2	**(b)** any ratio	Variable.
3	**(b)** chemical reaction	Law.
4	**(d)** negative charge	Negative.
5	**(a)** well-tested explanation	Explains.
6	**(a)** filtration	Filter paper.
7	**(b)** cation	Cation.
8	**(a)** 2 electrons	2n squared.
9	**(a)** 2 km	Divide by 1000.
10	**(b)** mass of solute + mass of solvent	Sum.
11	**(b)** ionic	Transfer.
12	**(b)** atomic number	Z.
13	**(d)** metre	m.
14	**(b)** mixture of gases	Homogeneous mixture.
15	**(a)** Lavoisier	Lavoisier.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Simplified representation of a real system (2).	2
17	Substance made of two or more components not chemically combined (2).	2
18	Mass is neither created nor destroyed in a chemical reaction (2).	2
19	Electron (-), proton (+) (1); Neutron (0) (1).	1+1
20	Approximate calculation (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Length: metre (1); Mass: kilogram (1); Time: second (1).	1+1+1
22	Clear vs turbid (1); Stable vs settles (1); Cannot be filtered vs can be filtered (1).	1+1+1
23	Matter has atoms (1); Atoms of an element are identical (1); Atoms combine in whole-number ratios (1).	1+1+1
24	Most alpha particles passed (1); Some were deflected (1); Few rebounded (1).	1+1+1
	OR: Burning (1); Rusting (1); Cooking food (1).	1+1+1
25	Solution = 200 g (1); 20/200 x 100 (1); 10 per cent (1).	1+1+1
	OR: Low melting point (1); Do not conduct (1); Soluble in organic solvents (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Question (1); Hypothesis (1); Experiment (1); Observation (1); Conclusion (1).	1+1+1+1+1
	OR: Homogeneous (1); Heterogeneous (1); Solution examples (1); Suspension examples (1); Colloids (1).	1+1+1+1+1
27	Statement (1); Reactants and products (1); Burning of carbon (1); 12 g + 32 g = 44 g (1); Closed system (1).	1+1+1+1+1
	OR: Positive sphere (1); Electrons embedded (1); Plum pudding (1); Neutral atom (1); Drawback (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** positive charge	Positive.
2	**(d)** a pattern	Regularity.
3	**(b)** evaporation	Water evaporates.
4	**(a)** anion	Anion.
5	**(d)** 8 electrons	2 x 4.
6	**(c)** 0.5 kg	Divide by 1000.
7	**(b)** 100 g	Sum.
8	**(c)** covalent	Sharing.
9	**(a)** mass number	A.
10	**(a)** kilogram	kg.
11	**(a)** homogeneous mixture	Uniform.
12	**(d)** 1 : 8	By mass.
13	**(b)** no charge	Neutral.
14	**(a)** approximate value	Approximate.
15	**(c)** sublimation	Camphor sublimes.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Filtration (1); Distillation (1).	1+1
17	A charged atom or group of atoms (2).	2
18	Combining capacity of an element (2).	2
19	Study of matter and its changes (2).	2
20	Solid changes directly to vapour (1); Camphor (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Be 2, 2 (1); B 2, 3 (1); C 2, 4 (1).	1+1+1
22	Reliability (1); Reduce errors (1); Check the results (1).	1+1+1
23	Composition (1); Properties (1); Separation (1).	1+1+1
24	Gain electrons (1); Cl^- (1); O^{2-} (1).	1+1+1
	OR: Medicine (1); Nuclear energy (1); Research (1).	1+1+1
25	54 x 5/18 (1); = 15 (1); m/s (1).	1+1+1
	OR: Milk (1); Fog (1); Butter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Physical: no new substance (1); Chemical: new substance (1); Examples of physical (1); Examples of chemical (1); Signs (1).	1+1+1+1+1
	OR: Homogeneous (1); Small particles (1); Stable (1); Transparent (1); No Tyndall effect (1).	1+1+1+1+1
27	NaCl (1); MgO (1); CaCl2 (1); Al2O3 (1); H2O (1).	1+1+1+1+1
	OR: Definition (1); Hydrogen (1); Chlorine (1); Uses (1); Same chemical properties (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Na^+	Loses 1 electron.
2	**(a)** 2, 4	C.
3	**(a)** 2 minutes	60 s per minute.
4	**(c)** 10 per cent	10/100.
5	**(c)** solids with high melting point	Crystalline.
6	**(a)** three isotopes	Protium, deuterium, tritium.
7	**(d)** second	s.
8	**(b)** suspension	Settles.
9	**(c)** matter	Matter.
10	**(c)** protons and neutrons	Nucleons.
11	**(a)** c	c.
12	**(a)** magnet	Magnetic.
13	**(c)** Cl^-	Gains 1 electron.
14	**(b)** 2, 6	O.
15	**(a)** temperature	Temperature.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Ionic bond (2).	2
17	Z = 11 (1); 2, 8, 1 (1).	1+1
18	Metre (1); Kilogram (1).	1+1
19	Homogeneous mixture of two or more substances (2).	2
20	A compound has a fixed ratio of elements by mass (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	12 + 32 (1); 44 g (1); Law of conservation of mass (1).	1+1+1
22	Atom is mostly empty (1); Nucleus is positive (1); Nucleus is small and dense (1).	1+1+1
23	24 x 60 (1); = 1440 (1); minutes (1).	1+1+1
24	5/100 x 200 (1); = 10 g (1); Water = 190 g (1).	1+1+1
	OR: 1, 3, 4 (1); 8, 2 (1); 5 (1).	1+1+1
25	F 2, 7 (1); Ne 2, 8 (1); Na 2, 8, 1 (1).	1+1+1
	OR: Parallax (1); Faulty instrument (1); Careless reading (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Atomic model (1); Globe (1); Ball and stick (1); Simplify reality (1); Change with evidence (1).	1+1+1+1+1
	OR: Magnet (1); Water (1); Filter (1); Evaporate (1); Draw conclusion (1).	1+1+1+1+1
27	Statement (1); Water ratio (1); 1 : 8 (1); Any source of water (1); Not for mixtures (1).	1+1+1+1+1
	OR: Nucleus (1); Electrons revolve (1); Empty space (1); Positive nucleus (1); Drawback (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.